Predicting myopia in medical students

Choosing models with 132 students: what worked, and how much of it to trust

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Machine Learning course, 2026/27

IN SHORT

The talk in three ideas

1

Simple is hard to beat

A logistic regression on the length of the eye is hard to beat with more complex models.

2

Medicine helped

A ratio from the medical literature gave a new model that passed our test on every target.

3

We measured our doubt

Repeating the whole selection shows that part of the gain does not survive.

THE PROBLEM

One dataset, four questions about myopia

M

Does the student have myopia?

SI / NO

MM

Myopia magna, over 6 dioptres?

SI / NO

Combo

Control, myopia or myopia magna?

C / M / MM

DCombo

Same, with myopia split in two

C / M1 / M2 / MM

Goal: labels for 24 test students with no answers, as accurate as possible, with a process we can defend.

THE DATA

Few students, few cases, and messy columns

132

training students, and 24 for the test

10

with MM = SI: always saying NO scores 0.92

49 to 33

raw columns to numeric features, each one justified

Phototype: a mixed-up column

Letters from a neighbouring column were typed in it, so 45 % of the training values are missing once they are removed.

Eight contradictory students

Combo = MM but MM = NO. We tried correcting them and kept the labels as provided: only the drop in accuracy was firm.

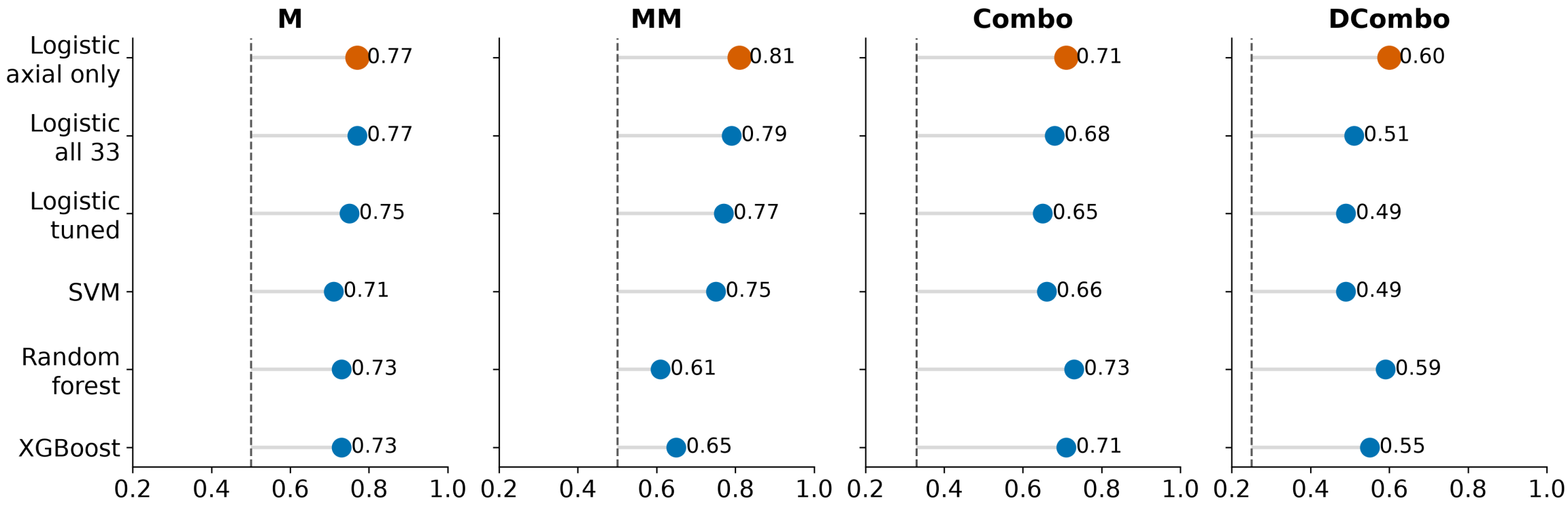
THE METHOD

With so little data, the method matters most

1. **Repeated CV** — 5 folds, 10 repeats. MM has exactly 2 positives per fold, so one split would mean little.
2. **No leakage** — Medians, scaling and tuning are learned inside the training folds, with nested CV.
3. **Written test** — A model must clearly beat the baseline on balanced accuracy without being clearly worse on macro-F1, or pass a second route, added after seeing validation results.
4. **One test use** — The 24 test students are used once, to predict, and nothing is chosen from them.

MODELS

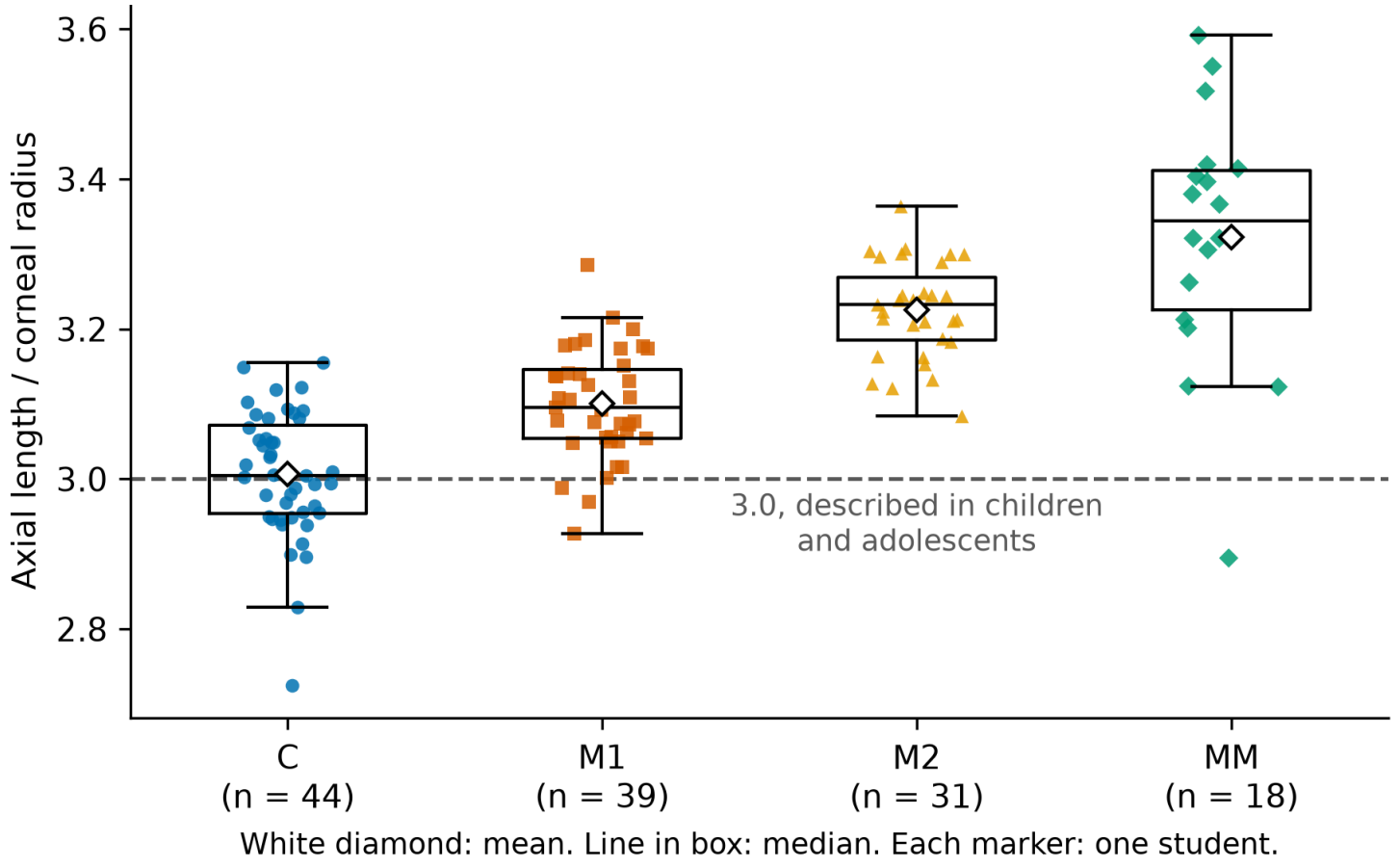
Complex models rarely beat a one-feature baseline



Mean balanced accuracy in cross-validation (dashed line: majority class)

XGBoost, new to us, builds small trees one after another, each correcting the errors of the previous ones. It never came first. Only the random forest on Combo is clearly above axial length alone.

Medicine gave us a better measurement



Axial length / corneal radius

The eye focuses through its length and its cornea together.

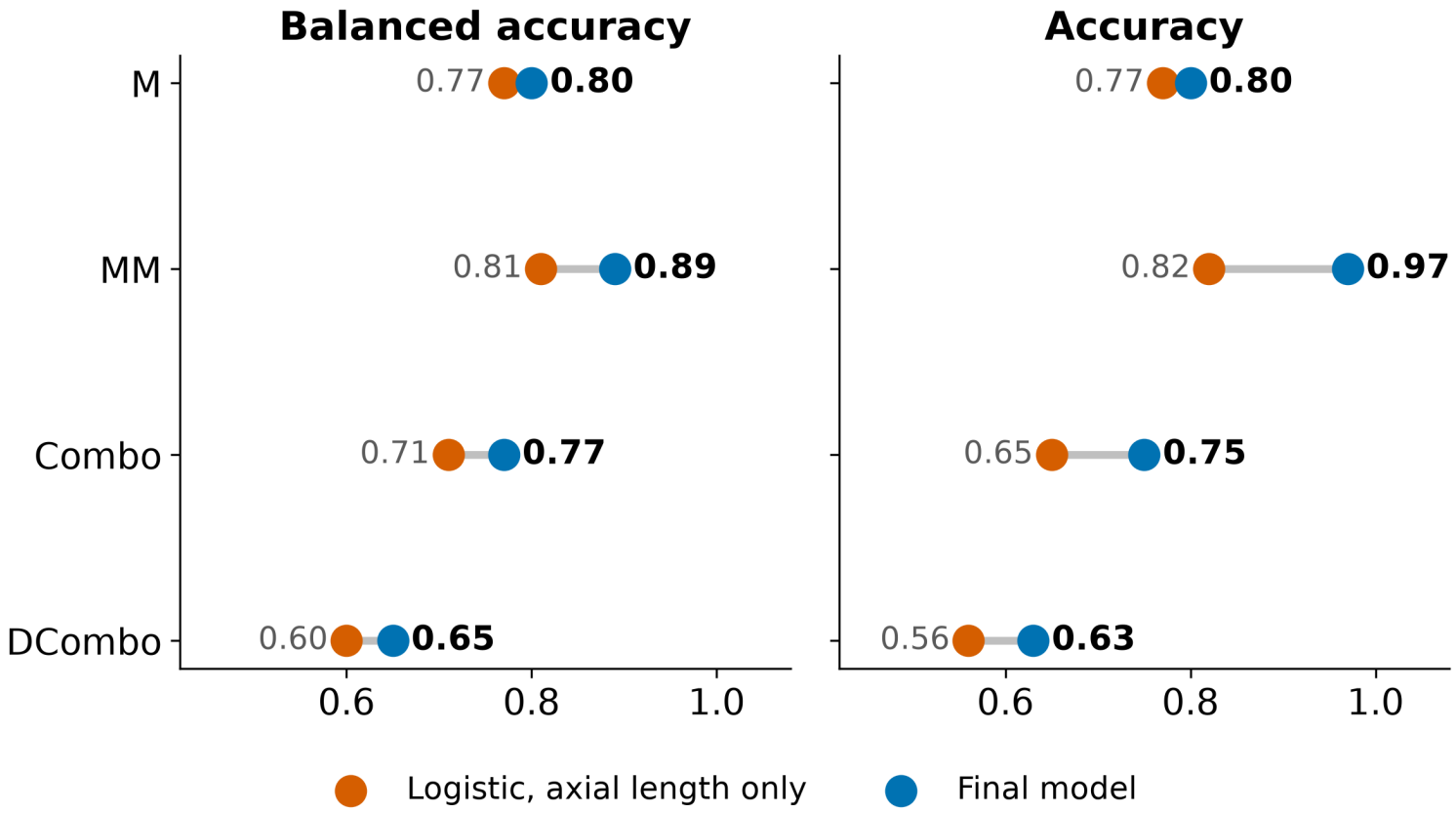
Computed for each student, without labels.

Units not stated in the brief: we deduced dioptres from the data.

Yebra-Pimentel et al., Arch Soc Esp Oftalmol 2004 (193 university students); Fan, Huang and Huang, J Ophthalmol 2024.

RESULTS

With the ratio, a new model won on every target



Final models

M, DCombo: logistic, axial + ratio.

Combo: random forest + ratio.

MM: logistic, ratio, tuned threshold.

Validation scores, before correcting for selection. An ordinal model for DCombo was tried and scored lower.

RESULTS

On MM, accuracy has a price: missed cases

Threshold 0.5

0.88

accuracy

0.90

of positives found (recall)

about 14

false alarms in 122 students

Tuned threshold (chosen)

0.97

accuracy

0.79

of positives found (recall)

about 2

false alarms in 122 students

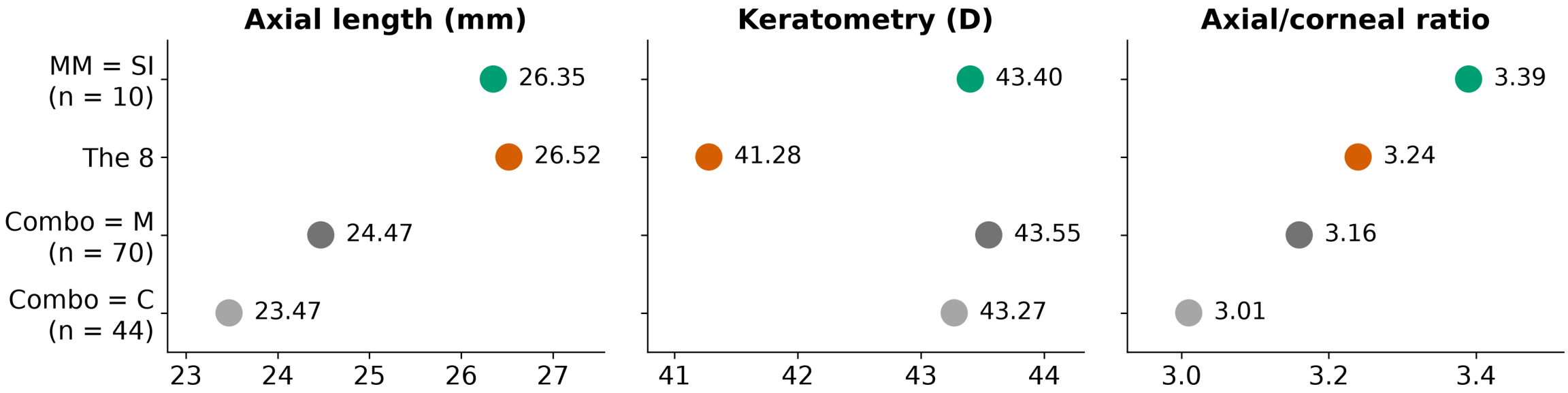
Always saying NO scores 0.92 accuracy and finds no one.

We kept the tuned threshold: it passed the second route of our test, and the test set is graded on accuracy.

In a clinic, the choice would depend on what each error costs.

RESULTS

The cornea may explain the eight odd students

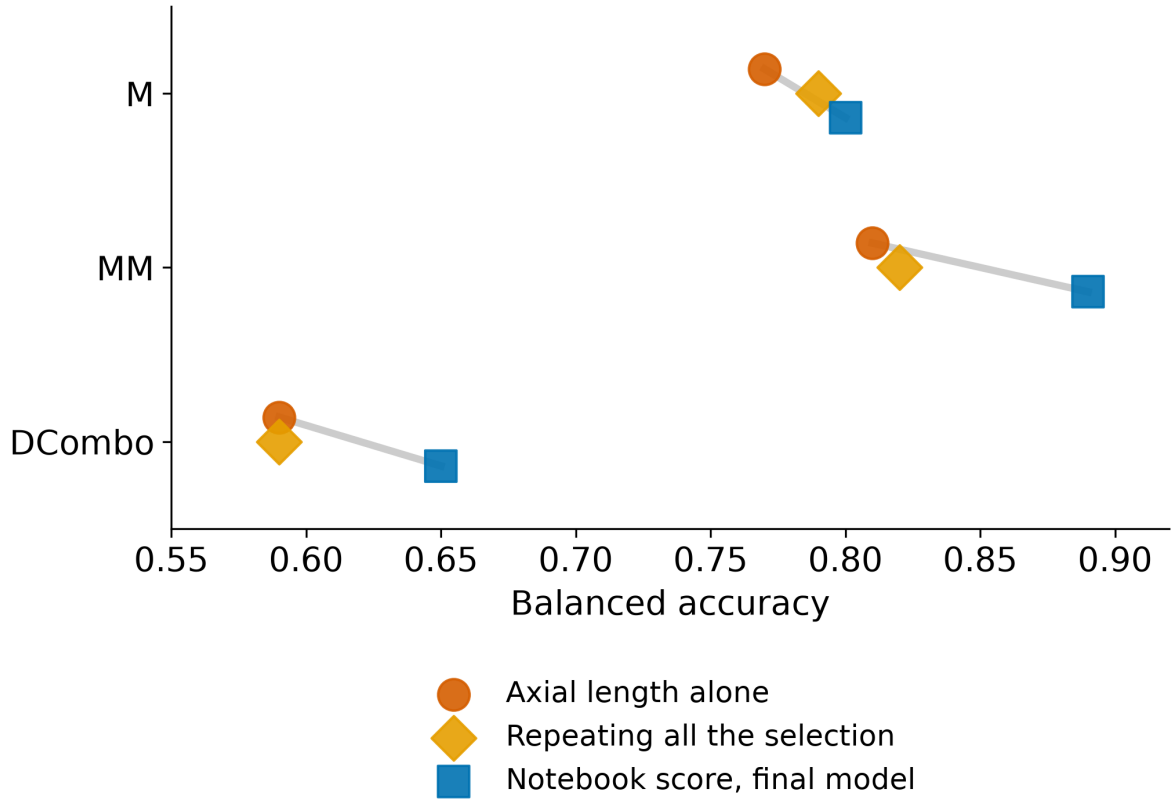


Same eye length as MM = SI, but flatter corneas. The cornea may partly compensate, keeping them under 6 dioptres.

A hypothesis: there are no refraction values in the data, and their ratios overlap with the MM = SI group.

HOW FAR TO TRUST IT

We measured how optimistic our scores are



Approximation with fewer students in training. Combo is left out: it uses a simpler forest than the chosen one.

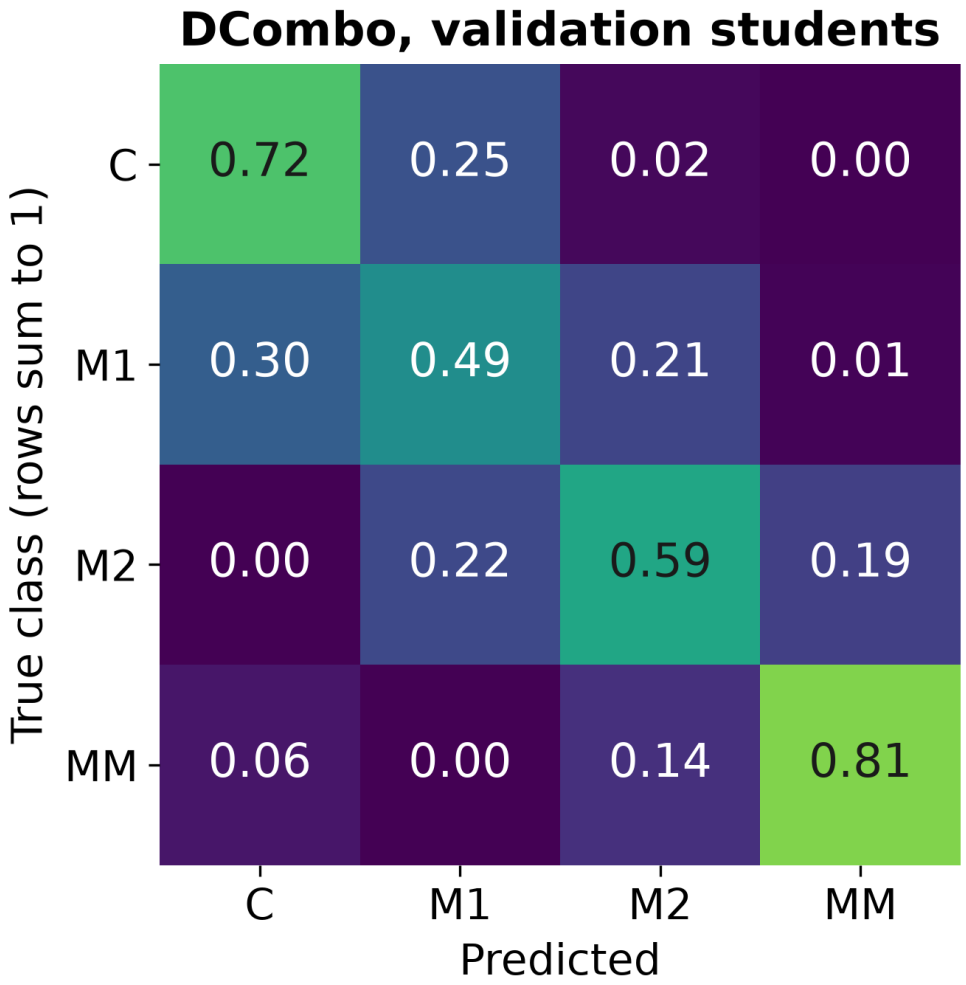
1 in 3

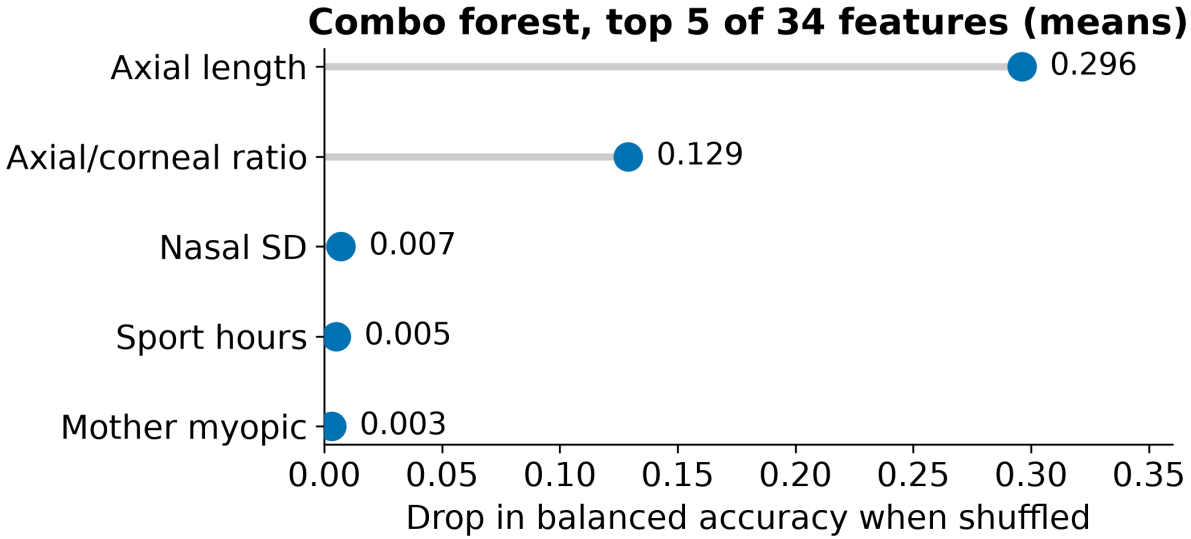
A useless feature passes our test on Combo about this often.

42

selection tests on the same 132 students. The cornea seems to carry real information; the exact gains are uncertain.

Two features drive the models; errors stay close



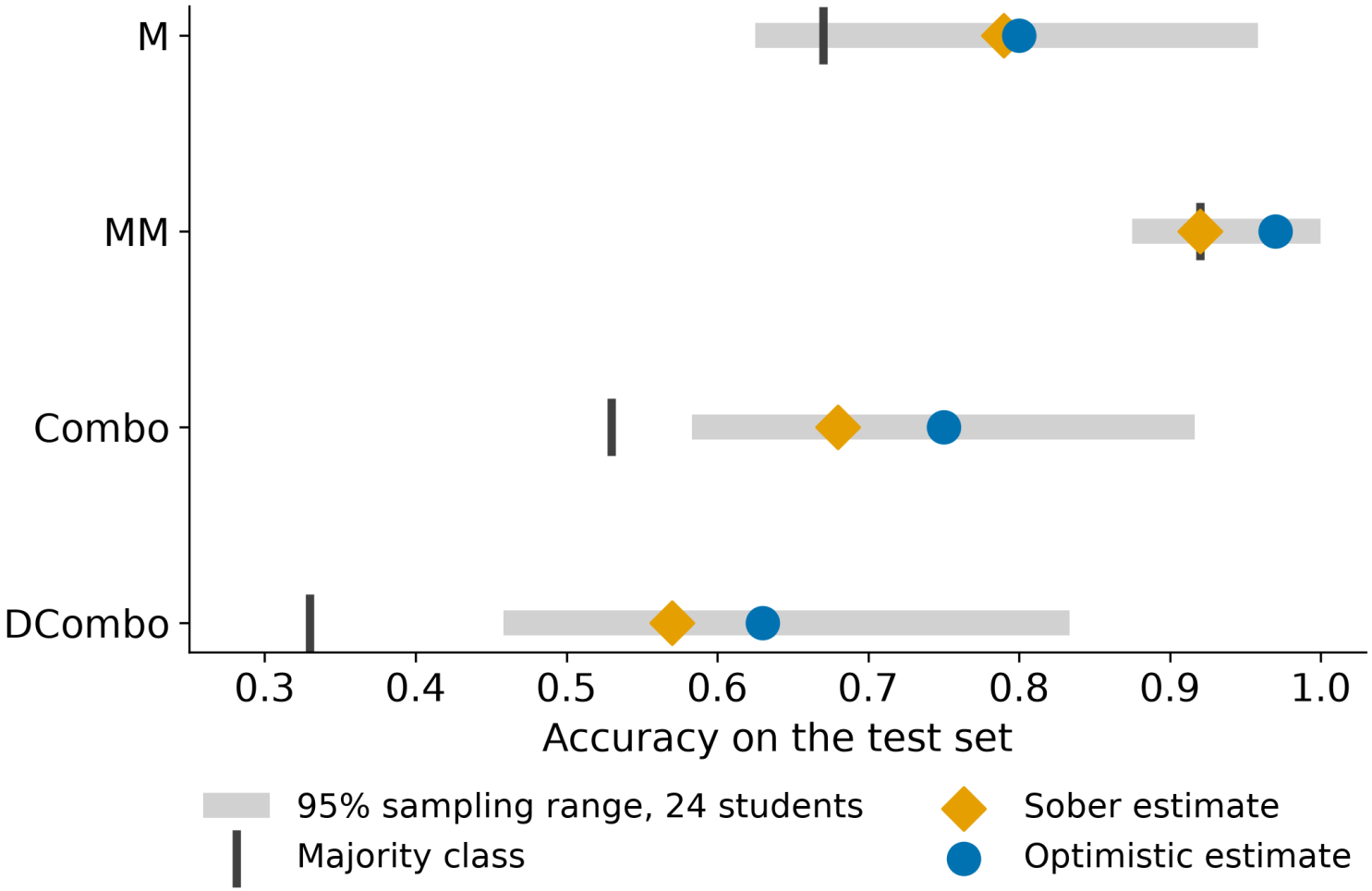


63 / 35 / 2

DCombo, %: correct, off by one class, off by more

This describes what the models use, not the causes of myopia.

What we expect on the test set



5 of 24

Test students predicted MM = SI, against 8 % in training.

Not changed: that would be choosing with the test set.

One student is worth about 0.04. The Combo sober estimate uses a simpler forest.

CONCLUSIONS

What we take away

1

Domain knowledge helped more than complexity

In our comparisons, a ratio from the literature did what added model complexity did not.

2

Measuring doubt is part of the result

A nested estimate and a useless-feature check show how far to trust the scores.

3

The limit is partly the data

More students and a direct measure of refraction would likely help more than another model.

REQUIRED BY THE BRIEF

Team and tools

Who did what

All parts were done together, taking the decisions jointly: Ana Gomez and Tomás Oriol Freixa.

Workload

50/50

Split evenly between the two members.

Keras / PyTorch

Neither

We used scikit-learn and XGBoost. With 132 students, neural networks were hard to justify.

REQUIRED BY THE BRIEF

How we used generative AI

What the AI did (Claude Code)

Generated much of the code, and drafted the notebook text.

Answered doubts and conceptual questions.

Proposed approaches, candidate models and selection rules, and found the literature on the ratio.

Reviewed methodology, charts and consistency with review agents.

Built these slides and drafted the video script.

What we did

Took the decisions, such as keeping the labels and accepting the rules.

Proposed ideas to the AI about what could be done.

Asked for the reviews and decided what to change.

Kept a log of the AI use, in docs/ai_usage.md.